

Student Academic Performance & Attendance Dataset

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Role	Data Analyst Intern
Subject	Statistical Analysis — Task 5
Tools Used	Python · Pandas · NumPy · SciPy
Report Date	June 15, 2026
Dataset	Student Academic Performance & Attendance Dataset
Records	1,005 Students

Executive Summary

This report presents a comprehensive statistical analysis of academic performance and attendance patterns across 1,005 student records. Using Python (Pandas, NumPy, SciPy), the analysis covers descriptive statistics across three core subjects — Mathematics, English, and Science — and applies hypothesis testing to explore the relationship between attendance rates and academic outcomes.

TOTAL STUDENTS	MATH MEAN	ENGLISH MEAN	SCIENCE MEAN	AVG. ATTEND.
1,005	75.20	70.87	78.02	84.89%
Dataset Records	Score / 105.38 Max	Score / 107.42 Max	Score / 102.43 Max	Std Dev ±5.18

1. Dataset Overview

The dataset comprises academic records for 1,005 students covering three subjects (Mathematics, English, Science) along with attendance data. All columns were verified for completeness — no missing values were detected — ensuring analytical integrity.

Attribute	Details
Source File	student_academic_performance.csv
Total Records	1,005 student entries
Features	Math_Score, English_Score, Science_Score, Attendance (%)
Missing Values	None detected
Data Types	Numerical (float64) for all score and attendance columns
Analysis Tools	Python 3 · Pandas · NumPy · SciPy

2. Descriptive Statistics

Descriptive statistics summarise the central tendency, spread, and range of each numerical variable, providing an immediate quantitative snapshot of student performance and attendance levels.

Metric	Math Score	English Score	Science Score	Attendance (%)
Mean	75.20	70.87	78.02	84.89
Median	75.26	70.76	77.95	84.98
Std Dev	9.83	12.06	7.90	5.18
Minimum	45.06	34.33	53.61	68.82
Maximum	105.38	107.42	102.43	100.90

Table 1 — Descriptive Statistics Summary (N = 1,005)

Key Observations

- **Science Scores are highest on average:** Mean of 78.02 with the lowest standard deviation (7.90), indicating consistent performance across students.
- **English Scores show the most variability:** Standard deviation of 12.06 — nearly 50% higher than Science — suggesting uneven student proficiency.

■ **Math Scores are centrally balanced:** Mean (75.20) $\approx$ Median (75.26), confirming an approximately symmetric distribution with minimal skewness.

■ **Attendance levels are strong overall:** Average attendance of 84.89% with a tight spread ($\pm 5.18\%$), indicating a disciplined student population.

■ **No ceiling effects observed:** Maximum scores (Math: 105.38; English: 107.42) are above 100, suggesting extra-credit or scaled scoring.

3. Mean, Median & Mode Analysis

Central tendency measures were calculated for the Math Score column to characterise typical student performance and assess distributional symmetry.

```
scores = df["Math_Score"]
print("Mean =", scores.mean()) # 75.20
print("Median =", scores.median()) # 75.26
print("Mode =", scores.mode()[0]) # 105.38
```

Measure	Value	Interpretation
Mean	75.20	Average Math performance across all 1,005 students
Median	75.26	Middle value; near-equal to mean → symmetric distribution
Mode	105.38	Most frequently occurring score; reflects high-achiever cluster

Table 2 — Central Tendency Measures for Math Score

Since Mean $\approx$ Median (difference < 0.1), the distribution of Math Scores is approximately symmetric with no significant skew. The high Mode value (105.38) suggests a secondary cluster of high-performing students, though it does not materially distort the overall distribution.

4. Hypothesis Testing — Independent Samples T-Test

An Independent Samples T-Test was conducted to determine whether a statistically significant difference exists in Math Scores between students with **high attendance ($\geq 85\%$)** and those with **low attendance ($< 85\%$)**.

Hypothesis	Statement
H_0 (Null)	No significant difference in Math Scores between high- and low-attendance groups
H_1 (Alternative)	A significant difference in Math Scores exists between the two attendance groups
α (Significance Level)	0.05 (95% confidence)

Python Implementation

```
from scipy.stats import ttest_ind

high_attendance = df[df['Attendance'] >= 85]['Math_Score']
low_attendance = df[df['Attendance'] < 85]['Math_Score']
```

```
t_statistic, p_value = ttest_ind(high_attendance, low_attendance)
print("T Statistic =", t_statistic) # -0.4265
print("P Value =", p_value) # 0.6698
```

T-Test Results

Parameter	Value	Threshold	Outcome
T-Statistic	-0.4265	—	Negative; high group slightly lower (negligible)
P-Value	0.6698	0.05	$p > \alpha \rightarrow$ Fail to Reject H_0
Decision	Fail to Reject Null Hypothesis		No significant difference detected
Confidence	95%	—	Result is statistically not significant

Table 3 — T-Test Results Summary

5. Statistical Interpretation

P-Value Analysis

The obtained p-value of 0.6698 is substantially greater than the significance level $\alpha = 0.05$. This means there is a 66.98% probability that the observed difference (if any) occurred due to random chance alone — far exceeding the acceptable 5% threshold.

T-Statistic Interpretation

The T-statistic of -0.4265 is very close to zero, indicating the group means are nearly identical. The negative sign simply reflects that the high-attendance group's mean is marginally lower than the low-attendance group — but the magnitude is too small to be meaningful.

Decision

We fail to reject the Null Hypothesis (H_0). There is no statistically significant difference in Math Scores between students with high ($\geq 85\%$) and low ($< 85\%$) attendance.

Practical Implication

While attendance is generally considered a positive academic indicator, this analysis suggests that Math Score outcomes in this dataset are not meaningfully driven by attendance rates alone. Other factors (study habits, prior knowledge, teaching methods) likely play a more decisive role.

6. Conclusions & Final Outcomes

The statistical analysis successfully addressed all objectives of Task 5. The findings are summarised below:

1. Descriptive Statistics Computed

Mean, Median, Mode, Std Dev, Min, and Max were derived for all three subjects and attendance, revealing Science as the strongest-performing subject and English as the most variable.

2. Distribution Symmetry Confirmed

Math Score distribution is approximately symmetric (Mean $\approx$ Median), validating the appropriateness of parametric testing.

3. T-Test Applied Correctly

An independent-samples T-Test was correctly applied to compare two distinct student groups (high vs. low attendance), using `scipy.stats.ttest_ind`.

4. Null Hypothesis Retained

With $p = 0.6698 > 0.05$, the null hypothesis is retained. No significant attendance-driven difference in Math performance was identified.

5. Analytical Pipeline Complete

The full pipeline — data loading, inspection, descriptive analysis, and hypothesis testing — was executed cleanly using Python, Pandas, NumPy, and SciPy.

7. Methodology Summary

Step	Activity	Tool / Method
1	Data Loading	pd.read_csv()
2	Data Inspection	df.shape, df.info()
3	Missing Value Check	df.isnull().sum()
4	Descriptive Statistics	df.describe()
5	Central Tendency (Math)	mean(), median(), mode()
6	Group Segmentation (Attendance)	Boolean indexing on df['Attendance']
7	Hypothesis Testing	scipy.stats.ttest_ind()
8	Result Interpretation	p-value vs. $\alpha = 0.05$

Table 4 — Analytical Methodology Steps

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Statistical Analysis — Task 5

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